

Cambridge IGCSE Mathematics

Extended Practice Paper

Cambridge IGCSE Mathematics 0580 Extended Mathematics – Grade 10 Practice Paper

Time: 1 hour 30 minutes

Total marks: 60

Instructions

- Answer all questions.
- Show all necessary working.
- Give non-exact numerical answers correct to 3 significant figures unless otherwise stated.
- A calculator may be used.
- Diagrams are not necessarily drawn to scale.

Question 1 – Number [6]

(a) Express 0.000384 in standard form. [1]

(b) Find the highest common factor of 84 and 126. [1]

(c) Simplify

$$\frac{3}{\sqrt{5} + 1}.$$

Give your answer in the form $a\sqrt{5} + b$, where a and b are rational numbers. [2]

(d) A quantity P is increased by 12% and then decreased by 12%.

Determine whether the final value is equal to the original value. Justify your answer.

[2]

Question 2 – Algebra

[8]

- (a) Expand and simplify

$$(2x - 3)(x + 5).$$

[2]

- (b) Factorise completely

$$6x^2 - 13x - 5.$$

[2]

- (c) Solve

$$\frac{2x - 1}{3} + \frac{x + 4}{2} = 7.$$

[2]

- (d) Solve

$$x^2 - 7x + 10 = 0.$$

[2]

Question 3 – Simultaneous equations and sequences

[7]

- (a) Solve the simultaneous equations

$$2x + 3y = 17,$$

$$4x - y = 11.$$

[3]

- (b) The first four terms of a sequence are

$$5, \quad 9, \quad 13, \quad 17.$$

- (i) Write down the next term.

[1]

- (ii) Find an expression for the n th term.

[2]

- (iii) Determine which term of the sequence is 101.

[1]

Question 4 – Coordinate geometry

[7]

The points $A(-2, 5)$ and $B(6, -3)$ are shown on a coordinate grid.

(a) Find the gradient of AB .

[2]

(b) Find the midpoint of AB .

[2]

(c) Find the equation of the straight line passing through A and B .

[2]

(d) A third point C has coordinates $(4, 7)$.

Determine whether AC is perpendicular to AB . You must justify your answer.

[1]

Question 5 – Geometry and mensuration

[8]

A sector of a circle has radius 8 cm and angle 135° .

(a) Calculate the length of the arc of the sector.

[2]

(b) Calculate the area of the sector.

[2]

(c) The sector is joined to a rectangle measuring 8 cm by 12 cm.
Calculate the total area of the resulting shape.

[2]

(d) Explain why the perimeter of the resulting shape cannot be found simply by adding the perimeter of the rectangle to the circumference of the full circle.

[2]

Question 6 – Trigonometry

[8]

A ladder of length 6.5 m rests against a vertical wall.

The bottom of the ladder is 2.8 m from the wall.

(a) Calculate the height reached by the ladder on the wall.

[3]

(b) Calculate the angle between the ladder and the ground.

[2]

(c) The ladder is moved so that its bottom is 3.5 m from the wall, while its length remains 6.5 m.
Calculate the new height reached by the ladder.

[3]

Question 7 – Probability

[7]

A box contains 5 red counters, 3 blue counters and 2 green counters.

Two counters are selected at random without replacement.

(a) Find the probability that the first counter selected is red. [1]

(b) Find the probability that both counters are red. [2]

(c) Find the probability that exactly one of the two counters is blue. [2]

(d) Find the probability that neither counter is green. [2]

Question 8 – Statistics

[9]

The times, in minutes, taken by 12 students to complete a mathematical task are

18 21 17 24 19 22 26 21 20 18 23 25

(a) Find the mean time. [2]

(b) Find the median time. [2]

(c) Find the range. [1]

(d) The student who took 26 minutes is removed from the data.
Explain whether the mean will increase, decrease, or remain unchanged. [2]

(e) Another student takes 35 minutes to complete the task.
Explain why the median is likely to be a better measure of the typical completion time than the mean. [2]

Question 9 – Functions and graphs

[8]

The function f is defined by

$$f(x) = x^2 - 4x + 1.$$

(a) Find $f(3)$.

[1]

(b) Solve

$$f(x) = 1.$$

[2]

(c) Write $f(x)$ in the form

$$(x - a)^2 + b.$$

[2]

(d) State the coordinates of the turning point of the graph of $y = f(x)$.

[1]

(e) Find the inverse function $f^{-1}(x)$.

State any restriction required for the inverse to be a function.

[2]

Question 10 – Problem solving

[10]

A company produces cylindrical containers.

Each container has radius 4 cm and height 12 cm.

The company wants to reduce the amount of material used while keeping the volume unchanged.

(a) Calculate the volume of the original container.

[2]

(b) The company proposes a new container with radius 6 cm.

Calculate the height required for the new container to have the same volume as the original.

[3]

(c) The surface area of a closed cylinder is given by

$$A = 2\pi r^2 + 2\pi rh.$$

Calculate the surface area of the original container.

[2]

(d) Calculate the surface area of the proposed new container.

[2]

(e) Determine which container uses less material. Give a reason for your answer.

[1]

End of Paper